Paras Rahul Pingale

AI / ML Engineer | Deep Learning & Generative AI | **Portfolio**

Pune, Maharashtra | +91 9518560080 | paraspingles@gmail.com | **LinkedIn** | **GitHub**

Professional Summary

Results-driven AI/ML Engineer and final-year B.E. student in Artificial Intelligence and Data Science. Possesses hands-on expertise spanning Deep Learning, Generative AI, LLMs, and end-to-end ML pipelines. Proficient in Python, PyTorch, and TensorFlow, with a demonstrated ability to build production-ready AI applications using RAG systems, AI agents, and CNN-based computer vision pipelines.

Technical Skills

- **Core Programming** Python (OOP, Decorators, Multithreading, Exception Handling), Java, C++, C.
- **Machine Learning** Scikit-learn, Linear & Logistic Regression, KNN, Decision Trees, Random Forest, Ensemble Models, Hyperparameter Tuning, Cross-Validation, Model Evaluation Metrics, Fine-Tuning / Optimization, Ensemble Learning, Confusion Matrix.
- **Deep Learning** PyTorch, TensorFlow, Keras, ANN, CNN, RNN / LSTM (NLP, Time Series), Transformers, Forward & Back Propagation, Gradient Descent, Dropout / Regularisation, Hyperparameter Optimization.
- **Generative AI & LLMs** Qwen2.5-3B, Llama 3, GPT, BERT, Hugging Face Transformers, RAG (Retrieval-Augmented Generation), Vector Databases, Prompt Engineering, AI Agents, Ollama, Gemini, API, API Integration.
- **Data Engineering** SQL, Pandas, NumPy, Data Cleaning & Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Matplotlib, Seaborn, Tableau, Statistical Analysis.
- **Computer Vision** OpenCV, CNN-based Detection & Classification, Image Preprocessing
- **Databases & Tools** MySQL, Jupyter Notebook, VS Code, Google Colab, Anaconda, Streamlit.
- **Automation** Python Automation, Email Automation (smtplib), File & OS-Level Automation, Scheduling, Logging, Automation Scripting.

Technical Projects

SmartHire – AI Mock Interview Agent Python · LLMs · Ollama · Llama 3 · Gemini API · Prompt Engineering · Streamlit **GitHub**

- Designed a multi-agent evaluation architecture where specialized agents independently assess technical accuracy, communication clarity, and confidence, then aggregate scores for more reliable, bias-resistant feedback.
- Integrated Ollama (Llama 3) and Gemini API with structured prompt engineering to generate role-specific interview questions and automated, personalized feedback reports.
- Built an automated PDF report generation pipeline, reducing manual evaluation effort by ~80% and enabling end-to-end candidate assessment
- Applied conversational AI design patterns to simulate realistic technical interview environments across multiple job roles.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain to retrieve relevant technical concepts grounding generated interview questions and feedback in target job description, reducing hallucinated or generic responses.

Text Sentiment Analysis (RNN / LSTM) Python · TensorFlow · Keras · NLTK · Word2Vec **GitHub**

- Built a stacked LSTM network for multi-class sentiment classification on movie-review and social-media text, with a full NLP pipeline (tokenization, stopword removal, Word2Vec embeddings).
- Applied dropout regularization and Adam optimizer tuning to control overfitting on variable-length sequences.

Surface Crack Detection using CNN Python · TensorFlow · Keras · OpenCV · Scikit-learn **GitHub**

- Built a binary CNN classifier for industrial surface-defect detection on a 40,000-image dataset, achieving **94.2% test accuracy**.
- Designed a multi-stage preprocessing pipeline (resizing, normalization, augmentation) and visualized confusion matrices, ROC curves, and activation maps to guide iterative model improvement.

Automated Machine Learning Algorithm Selection System Python · Scikit-learn · Pandas · NumPy · Matplotlib **GitHub**

- Built an AutoML framework benchmarking **7+ algorithms** across regression, classification, and clustering tasks, auto-selecting the best performer by Accuracy, F1, RMSE, and AUC.
- Automated hyperparameter tuning via GridSearchCV and result visualization, removing manual trial-and-error from model selection.

Plant Disease Prediction | Python · TensorFlow · Keras · Streamlit **GitHub**

- Developed a deep learning-based web application to identify 38 plant diseases across 14 crop species from leaf images using a CNN model.
- Trained and evaluated the model on 87,000+ augmented images, achieving 96.48% validation accuracy with optimized image preprocessing and regularization techniques.
- Built an interactive Streamlit interface enabling real-time image upload, disease classification, and instant prediction results for precision agriculture use cases.

Professional Experience

Web Development Executive (Intern) | Dialogues Media *Jun 2026 – Sept 2026*

- Engineered and deployed high-performance, responsive web applications leveraging **React, Vite, Tailwind CSS, and JavaScript**, delivering optimized user interfaces directly to production environments.
- Integrated server-side logic utilizing PHP to support dynamic front-end features, ensuring seamless data flow and application stability.
- Automated critical development workflows and streamlined the deployment lifecycle by implementing **CI/CD** pipelines and containerizing applications using **Docker**
- Shipped continuous **UI/UX improvements**, resolving frontend bottlenecks and enhancing overall end-user experience.

Achievements & Leadership

- **Team Leader, Smart India Hackathon (SIH) 2025:** Led team "404_Founder" through ideation and prototyping of an AI/ML solution, qualifying Round 1.
- **Innoverse Hackathon (Oct 2025):** Advanced to Round 2 for rapid prototyping under time pressure.
- **Team Lead, E-Cell,** PES Modern College — led ideation workshops & startup pitching | Social **Media Head,** Spandan Fest — digital outreach

Education

- **B.E. — Artificial Intelligence & Data Science (**2023 – Expected May 2027) Savitribai Phule Pune University
- **Secondary School Certificate (SSC) |** The Bishop Co-Ed School Undri, Pune | 2020 | **89%**

Certifications

Machine Learning with Python **freeCodeCamp** (Jul 2026) | Data Analysis with Python **freeCodeCamp** (Jul 2026) | Generative AI Workshop **Growthschool (**Feb 2026) | Advanced Green Skills & Artificial Intelligence **Edunet Foundation & AICTE** (Jan 2026)